

The Economic Impacts of Improving, Developing, and Maintaining Pennsylvania's
Transportation Infrastructure

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Abstract

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This paper serves as a tool for any potential readers to gain insight into the direct and indirect economic impacts of transportation infrastructure, and see how those impacts can affect Pennsylvania. The first chapter will deal with the term “public capital” and how it plays into the rest of the paper.

The second chapter of this paper will delve into the background of transportation infrastructure in Pennsylvania. Questions concerning how transportation infrastructure is funded will be addressed in this chapter. The term “public capital” will also be further deconstructed, and the economic implications that come with it will be analyzed.

The third chapter of this paper will be a literature review. The literature review will consist of what various academics, researchers and policy experts have to say about the economic impacts of transportation infrastructure. This will also entail seeking out alternatives to transportation infrastructure that are not already implemented.

The fourth chapter will use analyze data compiled from previous chapters, and ask various questions that will help tailor strong policy recommendations for the future. This will also include a comparative analysis of how other states fund their transportation infrastructure.

The fifth chapter and final chapter will summarize the paper's conclusions, discuss policy recommendations for state and federal policy makers regarding transportation infrastructure, and address gaps in the literature.

Chapter I: What is Transportation Infrastructure?

Transportation infrastructure is defined as: “...the physical facilities and systems that support economic activity” (ULI 2014). For the sake of this paper, the modes of transportation infrastructure that I will focus on are: roads, highways, railroads, and multimodal transportation (a combination of two or more means of transport). Transportation infrastructure in the United States has been characterized as “one gigantic assembly line for the production and transport of goods” (Block, *The Privatization of Roads & Highways*, 2009), which means that transportation infrastructure plays a very important role in the economic production process. This is due to the fact that transportation infrastructure is quite literally the medium through which capital moves. Without any way to reach centers of production (more commonly known as jobs), workers, materials and capital used for the production process would not be able to reach its fullest potential. Transportation infrastructure also plays an important role in the formation of human capital;¹ it has been found that “access to infrastructure provision promotes human development”.² A polity that lacks reliable transportation infrastructure can suffer from a lack of economic growth and capital accumulation.

To realize the importance of reliable transportation, look no further than Somalia, a war-torn country since 1991, located in the Horn of Africa. Somalia’s lack of central government and rule of law have led to a significant decline in its infrastructure. Somalia’s lack of reliable transportation infrastructure is an important element that has barred the meaningful creation of capital across the country. The mediums of transportation which America and many other developed countries take for granted; e.g. roads, railroads, highways, are severely lacking in

¹ Human Capital is defined as “the economic value of an employee’s skill set (Investopedia , 2016)

² (Kodongo & Ojah 2016).

Somalia. Imagine the road, railroad, or highway you use to go to work is unsafe and unreliable. This major impediment on a nation-wide scale would have adverse effects on a countries production output. Capital formation would be stunted, and the economy would be negatively affected. This case represents what could happen with the extreme negligence of transportation infrastructure, but it is not hard to imagine that continued disregard of transportation infrastructure in Pennsylvania can yield similar negative effects on its economy.

Purpose of the Paper

The purpose of this research paper is to better understand the role that transportation infrastructure has on the economy, specifically, in Pennsylvania. This research can be used to heighten the importance of the development and upkeep of transportation infrastructure in Pennsylvania and across the world, as well as establish a more concrete relationship between transportation infrastructure and capital, in its human or economic forms.

The second chapter of this paper will go over a number of key concepts that are pivotal to understanding the rest of the paper and its conclusions. The term “public capital” and its impacts on the economy will be examined. Chapter two will also look into the avenues of funding for infrastructure transportation in Pennsylvania.

The third chapter of this paper will be a literature review. The research of other academics and organizations will be analyzed, and the findings of the various pieces of literature that are most relevant to the topic will be applied. The findings of the literature review will then be summarized.

The fourth chapter of this paper will answer the overall question of “What is the impact that transportation infrastructure has on economies, and can have on Pennsylvania’s economy in

particular?” This will be done by utilizing the literature review from the third chapter, as well as checking the findings of various academics and organizations against the findings of Pierre-Richard Agenor, a professor who specializes in Macroeconomics and Policy Assessment. His findings on the economic impacts of public capital will be deconstructed in chapter two. This process will also include some comparative analysis between Pennsylvania and other states, in order to assess the most cost-effective ways to fund transportation infrastructure.

The fifth chapter will summarize the paper’s conclusions, and the policy implications of the research that has been conducted. There will be a discussion of recommendations for policy makers, and gaps in the literature on the topic will also be addressed.

Chapter II: Background

Background

Understanding Public Capital

Understanding public capital, its relationship to transportation infrastructure, and its impact on the economy is pivotal to this paper. Public capital is a form of capital that is commonly associated with infrastructure, and seen as an additional factor in the production process, along with labor and private capital.³ Public capital is the aggregate body of government-owned assets that are used as a means for private productivity. This includes highways, transit systems, roads and some railways. Public capital contributes to private productivity because it is a medium through which private capital is transported.

Public Capital may affect economic growth through three channels: a direct productivity

³ (Duran-Fernandez & Santos, 2014)

and cost effect on private production inputs, a complementarity effect on private investment, and a crowding-out effect on private spending through the financial system.⁴ It has been found that “...higher stocks of public capital would tend to raise the productivity of other inputs, such as labor and the stock of private capital, thereby reducing unit production costs.”⁵ In other words, the magnitude of this effect depends on the initial stock of public capital in the polity. It has been theorized that in mature economies with high stocks of public capital, marginal productivity⁶ effects from an increase in public capital are likely to be limited. However, in low-income countries where stocks of infrastructure assets are relatively low to begin with, the effects on marginal productivity would be substantial. In short, in areas with already established stocks of public capital, the effects of increasing public capital stock are not as significant as the effects of increasing public capital stock in areas with non-established/less mature stocks of public capital.

It has also been found that investments in public infrastructure capital can have a “significant positive effect on private investment”⁷, and has a complementarity effect on private investment. This holds true for the opposite, as inadequate public infrastructure may adversely affect private investment. Public capital may affect private capital formation indirectly through changes in output and relative prices. Adequate public infrastructure can lower transportation costs for private businesses, and in turn private businesses choose areas with adequate public infrastructure.

⁴ (Agenor, 2013)

⁵(Agenor, 2013)

⁶ Marginal Product is defined as a “Change in output that results from changing the labor input by one unit, (with) all other factors remaining constant.” (Business Dictionary , n.d.)

⁷ (Agenor, 2013)

An increase in public capital in infrastructure can also have a crowding-out effect on private investment. This crowding out effect can have adverse effects on growth if the drop in private capital formation “persists over time”⁸. This crowding-out effect stems from a number of variables. The method through which government funds public infrastructure can have a crowding-out effect. Increases in taxes, higher domestic interest rates, or increased public borrowing may drive away private investors, as these methods of funding for public infrastructure can raise the cost of borrowing, negatively affect expected after-tax rates of return on private capital, and increase the perceived risk of default on government debt.⁹ These crowding-out effects tend to be short term, as the increase in public capital stock raises output growth in the longer-term.

Public capital funding in Pennsylvania

Public capital comes in a number of forms in Pennsylvania. Most roads and highways, as well some forms of rail transportation are considered public capital. Out of the state’s 120,039 miles of roads, 41,103 miles are publically owned by either The Pennsylvania Department of Transportation (PennDOT), the Pennsylvania turnpike or other state agencies. 78,935 miles are owned by local governments or Federal agencies. Another 28,221 miles of the state’s roads are part of the federal aid system; 7,217 are part of the National Highway system and 1,867 are part of the Interstate system.¹⁰ Pennsylvania owns and operates federally aided roads; however the Federal government pays 90% of the cost.¹¹ The federal government funds Pennsylvania

⁸ ibid

⁹ (Agenor, 2013)

¹⁰ (Pennsylvania Department of Transportation, 2016)

¹¹ (Federal Highway Administration , 2015)

highways through the Highway Trust Fund. According to the Congressional Budget Office:

“The federal government’s surface transportation programs are financed mostly through the Highway Trust Fund, an accounting mechanism in the federal budget that comprises two separate accounts, one for highways and one for mass transit. The trust fund records inflows from revenues collected through excise taxes on the sale of motor fuels, trucks and trailers, and truck tires; taxes on the use of certain kinds of vehicles; and interest credited to the fund.” (CBO 2014)

The federal government grants money designated for highway and mass transit to state and local governments. More than \$50 billion is spent on these grants for “surface transportation programs.”¹² The funds are raised through excise taxes on the sale of motor fuels, trucks and trailers, and truck tires, as well as from taxes on the use of certain kinds of vehicles. Through the Trust Fund, the Federal government brings in nearly \$34 billion in tax revenue, but spends about \$50 billion on infrastructure each year, which has caused deficits in the budget.¹³ One reason for this is the fact that the federal tax on a gallon of gas (\$.0184) has not been increased since 1993, while the average cost of a gallon of gas has increased from \$1.12 in 1993, to \$2.29 in 2017.¹⁴

In 2015, the Federal government passed the “Fixing America’s Surface Transportation Act” (FAST), which will reauthorize the federal highway and public transportation programs for fiscal years 2016-2020. This will also stabilize the Highway Trust Fund during the same time period. The FAST act will transfer resources from the (Federal) general fund to the Highway Trust Fund; \$51.9 billion for highways and \$18.1 billion for mass transit, in order to fund the program authorizations. However, it is speculated that without an increase in excise taxes or a new source of revenue, the Highway Trust Fund will exhaust its funds provided by the FAST

¹² (Congressional Budget Office, 2014)

¹³ (The Hill, 2017)

¹⁴ (American Society of Civil Engineers , 2014) & (AAA , 2017)

Act by the end of fiscal year 2020.¹⁵ It is also speculated that Congress will need to find \$100 billion in additional Highway Trust Fund revenue for fiscal years 2021-2025.

The Constitution of Pennsylvania specifies that:

“All proceeds from gasoline and other motor fuel excise taxes, motor vehicle registration fees and license taxes, operators’ license fees and other excise taxes imposed on products used in motor transportation after providing therefrom for (a) cost of administration and collection, (b) payment of obligations incurred in the construction and reconstruction of public highways and bridges shall be appropriated by the General Assembly to agencies of the State or political subdivisions thereof; and used solely for construction, reconstruction, maintenance and repair of and safety on public highways and bridges and costs and expenses incident thereto, and for the payment of obligations incurred for such purposes, and shall not be diverted by transfer or otherwise to any other purpose....”¹⁶

The State Constitution requires revenue from excise taxes and fees to be used solely for construction, reconstruction, maintenance and repair of public highways and bridges in the state.

Act 89 of 2013 in Pennsylvania has paved the way for steadier transportation infrastructure funding in the state. Across a period of 5 years, and by the fiscal year of 2017-2018, \$2.3 billion will be generated for the states’ highways and roads, bridges, public transit, local government and other modal infrastructure systems. Act 89 accomplishes this by eliminating the flat gas tax of 12 cents, and uncapping the wholesale oil franchise tax as to offset and make the 12 cent elimination revenue neutral. The five year phase out of the oil franchise tax is estimated to generate an additional \$1.85 billion per year.¹⁷ To state government employees, Act 89 is seen as having a “huge impact on mass transit”¹⁸. Vehicle registration and driver licensing fees will be increased over the next several years, as well as some traffic violation fines. The following chart shows the effects of Act 89’s gas tax hike compared to other

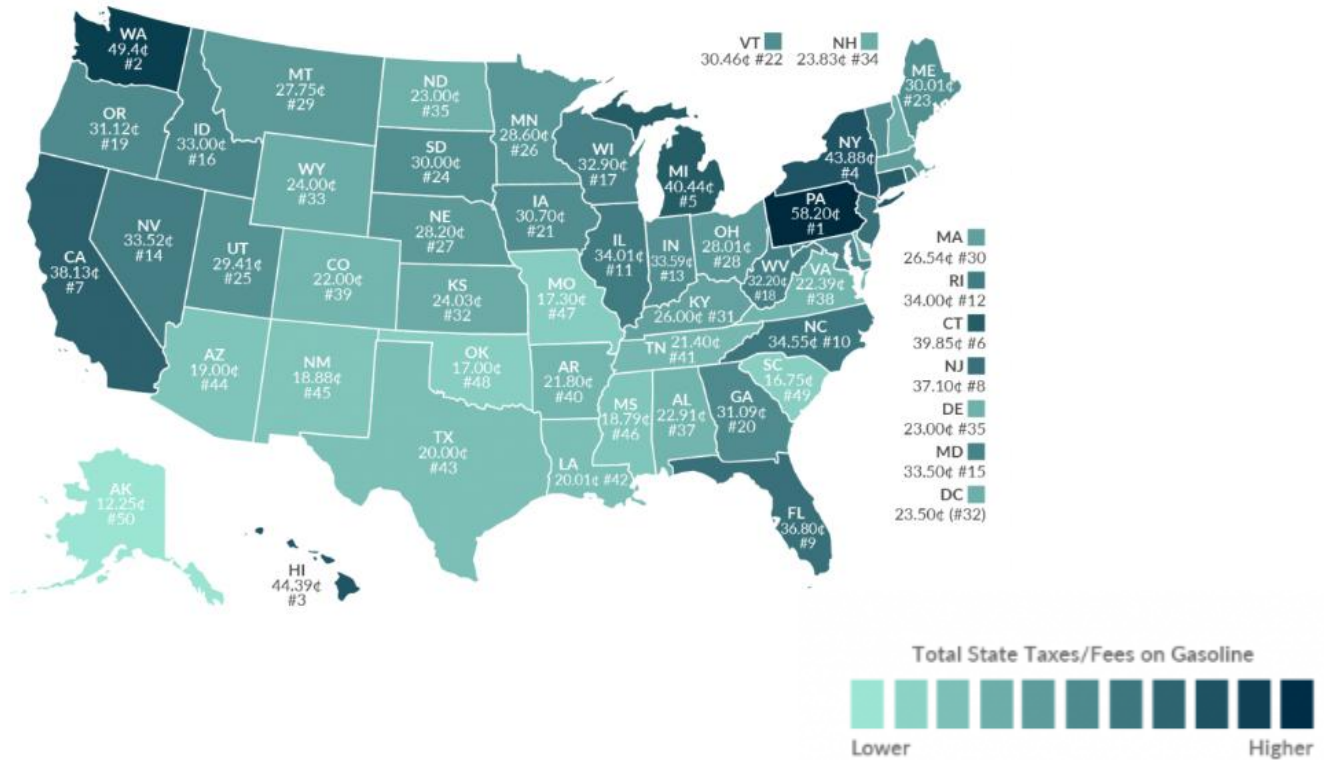
¹⁵ (American Road & Transportation Building Association , 2015)

¹⁶ 8 Pa. Cons. Stat. §11

¹⁷ (Associated Pennsylvania Constructors & Pennsylvania Highway Information Association , 2013)

¹⁸ (Meredith Biggica, 2017)

states in the nation:



19

In early 2017, Governor Tom Wolf announced a new program, named “Road Maintenance and Preservation.” Over a 10-year period from fiscal years 2018-19 to 2027-28, 4% additional revenue each year will be put towards PennDOT highway and bridge construction and county maintenance projects, totaling an estimated \$2.1 billion.²⁰ These funds are part of the State Police allocation from the Motor License Fund (\$802 million in 2017-18). Over the 10-

¹⁹ (Drenkard, 2017)

²⁰ (Pennsylvania Department of Transportation, 2017)

year-period, \$1 billion will be allocated to county maintenance, and \$1.1 billion will go toward highway and bridge capital projects.²¹ PennDOT states that this initiative will ensure that all maintenance projects that use Act 89 funds will be completed by 2028. This initiative will have a major impact on lower-volume routes in the more rural areas of Pennsylvania, as these roads may not be as much of a priority in other programs and bills that fund transportation infrastructure.

The program also makes strides towards environmentally friendly or green infrastructure, through the use of Recycled Asphalt Paving (RAP). RAP repurposes materials from projects onto other roadways by mixing ground millings with oil, which allows the state to pave less-traveled roads that otherwise would not be paved. This method saves the need to purchase new asphalt material, generating an estimated savings in one RAP region of \$5.4 million per year.²²

Passenger rail and Freight Rail

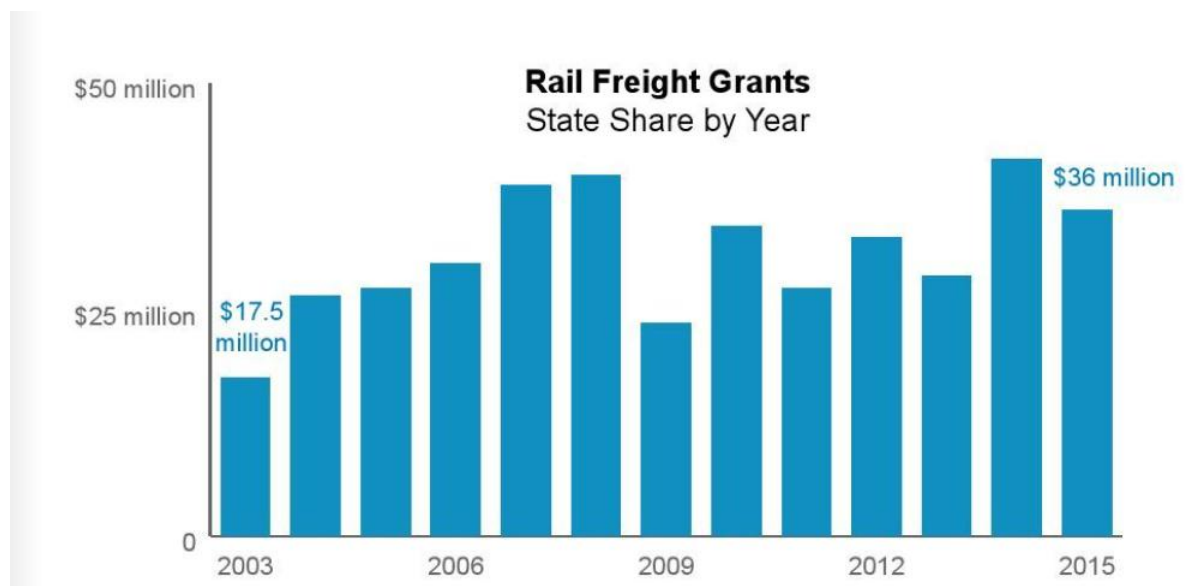
Passenger rail is funded by the federal government and PennDOT in Pennsylvania, and there are a number of passenger rail systems in Pennsylvania. Intercity rail (Amtrak), which is rail running between cities across the U.S., is funded by both the state and federal government. Commuter transit, which is short-haul rail passenger service operating in metropolitan and suburban areas, is provided by the Southeastern Pennsylvania Transportation Authority (SEPTA) in the Philadelphia metropolitan area. New Jersey transit also serves commuter rail into Philadelphia. The federal government funds railroad infrastructure improvements through Transportation Investment Generating Economic Recovery grants, the Federal Highway

²¹ Ibid.

²² Ibid

Administration, and the Federal Railroad Administration. The federal government also has a number of lending assistance and credit programs, including the Transportation Infrastructure Finance and Innovation Act (TIFIA) and the Railroad Rehabilitation and Improvement Financing Program (RRIF). Programs such as these provide loans to fund various transportation infrastructure projects which also extends to passenger rail and freight rail projects.

Pennsylvania is ranked 1st in the country in the number of operating railroads (65) and ranked near the top in total track mileage (over 5,600 miles).²³ Freight rail in Pennsylvania is not owned by the state, and therefore is not public capital, however the industry relies heavily on state and federal funding. The Rail Freight Assistance Program and Rail Transportation Assistance program are grant programs run by Pennsylvania to help fund rail freight in the state. The 2013 State Comprehensive Transportation Funding Plan (also known as Act 89) is a long-range state source of funding for freight rail as well. Money is set-aside for freight rail, beginning at \$8 million and is increased to \$10 million.



²³ (Pennsylvania Department of Transportation, 2016)

The Pennsylvania Infrastructure Bank is a loan fund administered by PennDOT, and offers flexible financing opportunities for transportation improvement projects throughout the state. The bank provides low-interest loans for infrastructure projects, and was capitalized with federal and state funds in 1998.²⁴

Private-Public partnerships, authorized by Act 88 of 2012, are collaborations between the public and private sectors. The Public-Private Transportation Partnership board guides these investments. Public-private partnerships do the following:²⁵

- Transfer the responsibility of a facility's engineering, construction, operation and/or maintenance to the private sector for a defined period of time;
- Allow the private sector to perform by contract a service previously provided by the public sector; and
- Ensure the private firm receives payments either from existing revenue sources or through the collection of new tolls or user fees.

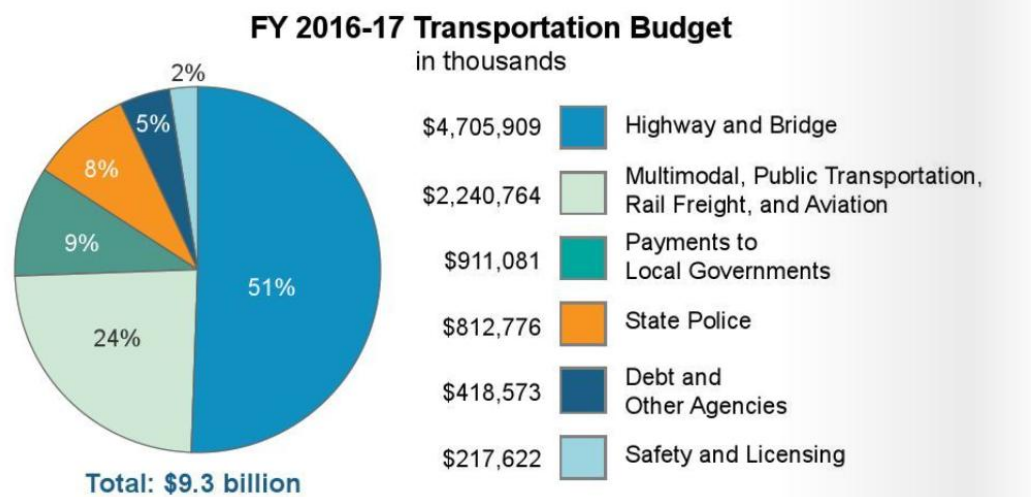
Basic Public-Private partnership projects either build new facilities or improve existing facilities.

According to the Pennsylvania State Transportation Commission, the state transportation budget for fiscal year 2016-2017 is approximately 12% more than the previous fiscal year.²⁶

²⁴ (Pennsylvania Department of Transportation, 2016)

²⁵ (Pennsylvania Department of Transportation , 2015)

²⁶ (State Transportation Commission , 2017)



Source: Pennsylvania State Transportation Commission

The Current State of Transportation Infrastructure in Pennsylvania

National infrastructure maintenance and development was a subject of discussion for candidates in the 2016 Presidential election. Both Hillary Clinton and Donald Trump could agree that the federal government should invest more in the state to make Pennsylvania transportation infrastructure projects a reality.²⁷ Both promised to “pump billions of federal dollars into infrastructure”²⁸ if elected, in order to upgrade the nation’s roads, bridges and rails. This would subsequently create millions of jobs. Federal dollars make up almost two-thirds of road funding and almost a quarter of transit funding. The fact that both candidates can agree on a boost of federal funding for Pennsylvania transportation infrastructure shows its importance.

For 2017, the American Society of Civil Engineers rated the United States as having an infrastructure report card grade of a D+. Bridges were graded a C+, public transit received a D-,

²⁷ (Laughlin, 2016)

²⁸ *ibid*

roads received a D, and rail received a B.²⁹ The latest ASCE report for Pennsylvania rates the state as having an overall grade of a C-, which means its infrastructure was mediocre.

Pennsylvania bridges were graded a D+, public transit received a D, roads received a D- and rail received a B.

Bridges

Bridges are rated a D+ or “poor”. Approximately 23% of Bridges in Pennsylvania were considered structurally deficient in 2014, and 19% were considered functionally obsolete. The implications of bridge closures or weight restrictions as a result of out-of-date bridges create traffic and congestion on both a local and regional level. This can impact the economy of the region as well. Even with Act 89 funding, the ACSE estimates that “more than 50 percent of the needs for state bridges and more than 60% of the funding needs for local bridges will still not be met in 2019” (American Society of Civil Engineers , 2014).

Transit

Transit is rated a D or “poor”. The use of Pennsylvania’s public transportation has increased more than 33% since 1995, with public transportation providers being present in every county in the state in one form or another. Existing transit systems in Pennsylvania have “a backlog of required reinvestment to replace aging infrastructure due to historically unstable funding coupled with rising costs and deferred maintenance” (American Society of Civil Engineers , 2014). Lack of funding will pose challenges for the future of transit in the state. While Act 89 provides funding stability for transit systems across the state, local funding sources should be utilized to increase capital expansion.

²⁹ (American Society of Civil Engineers, 2017)

Roads

Roads are rated a D or “poor”. Truck traffic on Pennsylvania roads averages over 34%, which is more than double the national average.³⁰ 44% of roads were rated fair or poor by PennDOT. ACSE estimates that without construction of new roadways and lanes to increase capacity, traffic congestion will increase and road quality will decline. Traffic congestions costs the average commuter 182 hours of delay and 86 gallons of fuel wasted every year in major urban areas.³¹ The statewide congestion is estimated to cost drivers over \$3.7 billion per year in lost time and wasted fuel. The ASCE estimates that ACT 89 spending for roads will not be enough, and that the roadway funding will not be met in 2019. More funding will be needed to keep the road system from degrading and improving it.

PennDOT gets more specific in which types of roads need maintenance and/or upgrading. Pennsylvania Interstates should be considered for reconstruction every 40 years, however 59% of them are older than 40 years.³² This means 1,600 segment miles in the state have not yet been considered for reconstruction when they should have been. Pennsylvania’s lower-volume roads comprise of 18,000 miles. Over 3,200 miles (18%) have not been seal coated in 7 or more years, and more than 4,300 miles (24%) have not been resurfaced in over 20 years. The suggested cycle to seal coat these roads is every 4-7 roads, and the suggested cycle to resurface is ever 15-20 years.³³

Rail

³⁰ (American Society of Civil Engineers , 2014)

³¹ *ibid*

³² (Pennsylvania Department of Transportation, 2017)

³³ *ibid*

Pennsylvania freight rail is mostly privately owned. Most railroads are generally able to finance their own capital investments, however, short line railroads tend to have difficulty in making infrastructure investments to remain viable and competitive. The ASCE reports that “projects that could be undertaken to address the Commonwealth’s infrastructure needs total more than \$280 million.” (American Society of Civil Engineers , 2014). Freight rail is rated the highest on the report card, and is considered “Good”.

It is apparent that Act 89 of 2013 has introduced well needed funding for transportation infrastructure. However, funding will not be sufficient for future infrastructure needs. The recently announced Maintenance and Preservation program is another essential funding source, however, neither Act 89 nor the Maintenance and Preservation program are enough. Other than rail freight, which is mostly privately owned but publically funded, transportation infrastructure needs improvement in Pennsylvania.

Chapter III: Literature Review

Transport Infrastructure and Real Estate

There is a wide array of literature pertaining to the economic impacts of transportation infrastructure. Through interviews and research, the Urban Land Institute concluded that transportation infrastructure is a “driver of real estate and development, especially by those who are in the business of providing it.”³⁴ According to a study by ULI, 88% of public and private leaders viewed infrastructure quality as the top factor that drives where real estate development

³⁴ (Urban Land Institute , 2014)

occurs. Public transit was seen as the highest infrastructure priority. Infrastructure beat out regulations and tax structure as the top factor driving real estate development. Infrastructure development helps to keep cities more competitive; the quality of infrastructure in a city can attract or repel real estate developers. More competitive cities will want to keep high-quality infrastructure in order to stand out against other cities.

ULI also found that the public's willingness and ability to pay for infrastructure is seen as the most important factor in the future of infrastructure and real estate over the next decade. It was found that cooperation between developers and local government, public-private partnerships, was identified by three-quarters of respondents as the most significant funding approach for new infrastructure over the next decade. ULI noted that "...when it comes to funding, many options need to be on the table."³⁵ Respondents of the survey felt that long-term operations and maintenance for infrastructure are usually neglected, with private sector respondents having a more negative take on the subject than public sector respondents.

ULI suggests that communities must "overcome reluctance to fund infrastructure at adequate levels..."³⁶ and that governments must commit to infrastructure projects for the long-term, as most infrastructure projects are long-term. In order to accomplish this, jurisdictions should communicate the benefits and economic returns of infrastructure investments to the public. By taking a look at the connection between infrastructure and real estate, governments will better be able to understand new ways to fund projects, as well as better ways to plan cities and prioritize projects.

The American Society of Civil Engineers theorizes that inefficient infrastructure will

³⁵ *ibid*

³⁶ (Urban Land Institute , 2014)

make businesses more inefficient. This will eventually lead to a drop in GDP, as well as reduce personal incomes. The average annual investment gap for surface transportation through 2025 is expected to increase from \$91 billion to \$110 billion. This is the gap between the amount of money that *should* be spent on infrastructure and the real amount of money that is being spent. The true costs and economic impacts of the deterioration of surface transportation infrastructure has not always been immediately apparent, because its deterioration has been diffused over time and throughout the nation.

The most affected regions of surface transportation infrastructure deterioration are those with the largest concentrations of urban areas. Urban highways, bridges and transit systems are “generally in worse condition today due to more congestion... (and) faster rates of deterioration” (American Society of Civil Engineers, 2016). The ASCE has concerns that the FAST act will barely keep up with inflation, and that the funding level is close to flat. The ASCE found that the poor conditions of transportation infrastructure will lead to:

“...increases in operating costs for trucks, cars, and rail vehicles... costs (to) damage to vehicles from deteriorated roadway surfaces, imposition to both additional miles traveled, time expended to avoid unusable or heavily congested roadways, or due to the breakdown of transit vehicles, and the added cost of repairing facilities after they have deteriorated.” (American Society of Civil Engineers, 2016)

Transportation Infrastructure and Credit Ratings

Research has found that the health of a states’ transportation infrastructure can affect its credit ratings. There are a number of ways that transportation infrastructure can convey information to a credit rating agency. The physical health of public infrastructure is closely

related to a bond issuer's economic base, as well as its future growth prospects. A well-developed and maintained public infrastructure system may be able to ease future fiscal pressure. The physical condition of public infrastructure reflects governments' management capacities and practices. The health of a state's transportation infrastructure has the potential to reflect its management skills and financial strength, as well as its potential economic growth.³⁷ A study found that "the overall health of state highway infrastructure does matter in the assignment of state credit ratings..." (Chen, A, & Wang, 2016). There are real financial costs to neglecting infrastructure.

International Examples

There is a wide array of literature on the economic impacts of investing in and developing transportation infrastructure in various African and Asian countries. Aforementioned in this paper, public infrastructure is perceived as "...an input in the production process."³⁸ By understanding public infrastructure, and in turn public capital, as an important factor in economic growth, leaders across the developing world have sought to increase public capital stock. Deficits in infrastructure has been seen as an "important binding constraint on some traditional antecedents of economic growth, in addition to the fact that it may also directly enable growth."³⁹ A study suggested that the public infrastructure stocks of most African countries are not sufficient enough to support their required levels of production.⁴⁰ The same study concluded that in the case of Sub-Saharan African countries, infrastructure access relates to economic

³⁷ (Chen, A, & Wang, 2016)

³⁸ (Kodongo & Ojah 2016)

³⁹ *ibid*

⁴⁰ *ibid*

growth indirectly through trade competitiveness and that infrastructure quality indirectly influences economic growth via “cross-border capital flows and export diversification.”⁴¹

An empirical study that looks at China from 1986 to 2011 found that economic impacts of improving transportation infrastructure differs region to region and at the national level. These differences are results of the phases of economic development, transportation investment policy, transportation infrastructure level, spillovers from other regions, and reform policies carried out by the central government.⁴² The study ultimately found that transportation investments have statistically significant impacts on economic growth.

In both cases, the economic impacts of transportation infrastructure investment are significant. As previously stated, places with initially low stocks of public capital receive more significant economic impacts from expanding their public capital stock, as demonstrated by Sub-Saharan Africa and China. Although Pennsylvania has a relatively high and developed stock of public capital compared to Africa and China in that aspect, it is important to understand the potential economic impacts that transportation infrastructure can have on a country.

Public vs. Private: Privatization as a solution?

There are several different definitions of “privatization” when it comes to roadways and other forms of transportation infrastructure. For the sake of this paper, the different kinds of privatization that previous literature has established will be defined as:

- Total privatization of roadways: Roads, highways, and bridges are totally under ownership of the private sector and subject to market forces; and

⁴¹ *ibid*

⁴² (Jiang, He, Zhang, Qin, & Shao, 2017)

- Public-Private Partnerships: The Government leases roads, bridges, or toll roads to the private sector for a set amount of time.

In many cases, people refer to Public-Private partnerships which are much more commonplace. The total privatization of roadways is different, as roadways would be totally subject to market principles and the private sector. In both cases, the tax burden on the public is lessened as less roads need to be funded by government taxes.

The total privatization of the nation's roads, highways, and bridges has been seen as a viable alternative by some economists.⁴³ Economist Walter Block perceives the current problems that face America's roadways as a result of mismanagement by bureaucrats. In turn, free-market transportation would see the profit-driven incentive to build and maintain roads. The "road entrepreneur" would have to "try to contain congestion, reduce traffic accidents, plan and design new facilities in coordination with already existing highways, as well as with the plans of other for new expansion."⁴⁴ In theory, market competition would incentivize road entrepreneurs to make their roads more competitive and attractive when compared to other roads, and therefore find solutions to the current problems that public roadways face. Block also addresses the concern of over-tolled private roadways. He states that roads with a high number of toll booths per mile would be "avoided like the plague by customers, who would happily patronize a road with fewer obstructions."⁴⁵

Pivotal to Blocks' argument is the notion that roadways are not "public goods"⁴⁶ and do

⁴³ (Block, The Privatization of Roads & Highways, 2009)

⁴⁴ (Block, 2009)

⁴⁵ *ibid*

⁴⁶ A public good is defined as "a product that one individual can consume without reducing its availability to another individual, and from which no one is excluded." (Investopedia, n.d.)

not necessarily need to be public capital. By seeing roadways as another private good just like food or cellphones, it is possible to see how roads can be run by the private sector. Private roads are already used, just not in mass scale; enclosed shopping centers are used as an example of modern day private roads. Railroads have also been run by “profit-making firms.”⁴⁷ Block identifies the most realistic method of privatizing the roads would either be to sell or give them away to those who live on them, work on them, or travel through them, or auction them off to the highest bidders.⁴⁸ To Block and economists like him, the public highway system is a “public firm that is large in size, lax in management, and a costly burden to taxpayers... suffering from problems of urban traffic congestion, poor maintenance, and high fatalities.”⁴⁹ Private roadways can provide solutions to these perceived problems, by spurring innovation as a result of the free-market, reducing congestion, and being a huge tax-saving for the public.

There are a number of detractors from the arguments that Block and several other economists present on the privatization of roads. The perceived dangers of road privatization are numerous. It has been argued that long-term road contracts pose a variety of serious threats to the public interest, as the public will lose control over transportation policy. The U.S. Public Interest Research Group (PIRG) stated that “Road privatization elsewhere has shown that a private operator’s profit motives lead to very different management decisions than a government would pursue.”⁵⁰

“Non-compete” clauses in privatization contracts limit the state’s ability to improve or expand publically owned roads, because improved public roads would compete with privately

⁴⁷ (Block, *The Privatization of Roads & Highways*, 2009)

⁴⁸ *ibid*

⁴⁹ *ibid*

⁵⁰ (Phineas Baxandall, 2007)

owned roads. It is argued that these clauses hinder the state's ability to conduct effective transportation policy. Toll road privatization can lead to private monopolies, because toll roads rarely compete with each other. Private toll decisions can also control traffic management, because raising or lowering toll prices will lead some people to choose different roads to get to their destination, which may "... clog local roads, increasing congestion and pollution in local communities."⁵¹ Privatization of roads can create a "tax" on normal policy making. Some privatization contracts require the state to pay investors compensation for reduced toll revenue when the state performs construction such as to add an exit or build a mass transit line down the median. This would add to the cost of construction, putting more financial strain on the taxpayer.⁵² Private roadways may not be able to guarantee state-of-the-art safety and maintenance standards. In order to cut operating costs, private companies may want to cut certain safety measures and maintenance costs. The fair price for a toll road is a "high stakes guessing game."⁵³ An upfront cost that a private company makes to purchase a road may pale in comparison to the amount of money it is worth in the long term.

It is important to differentiate between Walter Block's conception of the privatization of roads and the U.S. PIRGs' conception. Walter Block talks about the total denationalization of the roads, whereas the U.S. PIRG speaks more so on the process of government leasing roads to private companies for long periods of time, also known as public-private partnerships. However, it is still possible to analyze the topic using the arguments that Walter Block and the PIRG provide. The PIRG is concerned with public control of transportation policy, safety and

⁵¹ *ibid*

⁵² *ibid*

⁵³ (Phineas Baxandall, 2007)

maintenance, which are concerns that would be present with both the leasing of and the complete sale of roads to the private sector. The U.S. PIRG argues against road privatization because it considers roads a public good that should remain in public control, while Walter Block sees roads as any other commodity that can be bought and sold, and should be subject to market forces for improvement.

There have been serious considerations to privatize Pennsylvania roads (in a Public-Private Partnership), which came from former Governor Rendell in December, 2006. The Pennsylvania Turnpike would have been leased to the private sector for 75 years for a price of \$12.8 billion. According to the former Governor, the lease would have provided funds for roads, bridges and public transit systems, and that “Pennsylvanians will get more and pay less.”⁵⁴ An outspoken opponent to privatization, the then President and CEO of the American Trucking Association Bill Graves, stated that: “Private toll road operators need not be concerned about the social impacts of toll rates on low-income workers, or on the costs to businesses that depend on the highway for transporting employees, customers, goods or services.” (Carretta, 2008)

The Public-Private notion of Privatization is much more realistic (in terms of Public Policy trends) than the total-privatization of roadways. In both cases, the perceived economic benefits are less of a tax burden on the public, as well as a new source of revenue from the leasing/sale of public roads to the private sector. Subjecting roadways and other transportation

⁵⁴ (Carretta, 2008)

infrastructures to market forces such as market competition can lead to innovating ways of solving problems that riddle publically-owned roadways. However, public control of roadways has its perceived benefits as well, including price control of tolls, state-of-the-art safety and maintenance, and the ability to publically control transportation policy. Publically-owned toll roads are seen as valuable revenue streams for the government, and can be used to fund various projects within government. Toll roads are used as a revenue source for improving roads highways and bridges. They act as an excise tax, in the sense that the only people who are “taxed” are those who use the service (roads). Federal approval is needed to toll interstate highways, through the Federal Highway Administration.

Green Infrastructure

Transportation infrastructure can directly impact the environment through pollution. Transportation infrastructure, mostly in the form of roads and highways, collects a high concentration of contaminants, and storm water can wash these contaminants into streams, lakes, and bays.⁵⁵ Hard surface roads can concentrate rainfall during storms and empty the flow of water immediately via storm sewers into streams, rivers, and lakes. This can result into flooding, increased sedimentation and erosion, and the overall pollution of ecosystems. Green infrastructure includes various technologies engineered to absorb and filter excess runoff, instead of funneling runoff into large sewer pipes that directly pollute water sources.⁵⁶ There has not been a widespread implementation of these technologies, because of “the lack of data (or lack of awareness of the data) supporting the claims that these technologies control runoff and reduce

⁵⁵ (U.S. House of Representatives Committee on Science and Technology , 2007)

⁵⁶ Ibid

non-point source pollution.”⁵⁷ Another reason for a lack of implementation is resistance from builders to incorporate green transportation infrastructure into designs. This stems from a lack of understanding of the different options. Options for green transportation infrastructure designs depend on climate and build environment.⁵⁸ There are also financial costs to implementing green transportation infrastructure such as high costs associated with removing old materials and installing new surfaces, and the traffic that comes with roadwork would be extremely costly.⁵⁹ Possible methods to foster the development of green transportation infrastructure include increasing state and federal research on the subject, as well as using federal funding sources to incentivize the use of green infrastructure.

Possible Negative Impacts of Transportation Infrastructure

A study on the Istanbul metro states that the wider economic impacts of transportation infrastructure investments overlook the socio-economic inequalities it can create. Some discussions on the impacts of transport infrastructures suggest that they “are likely to form segments and pathways; enclaves and fragments which may create isolation and/or connectivity and fluidity.”⁶⁰ Transport systems, like the Istanbul metro, can facilitate integration, development and growth, while at the same time be the causes of disintegration, exclusion, and socio-spatial inequalities. The study found that while the transport system investment had positive economic impacts, the economic development around the metro stations led to issues of displacement and gentrification. Low-skilled labor workers are being spatially excluded from

⁵⁷ *ibid*

⁵⁸ *Ibid.*

⁵⁹ *Ibid.*

⁶⁰ (Beyazit, 2015)

their work places, as the Istanbul metro generated sectoral transformation; from industry and manufacturing related sectors to service based high-profile businesses within its catchment areas. The metro fostered high levels of capital investment and capital accumulation, which creates uneven geographies with other parts of the city. The metro is “likely to drive off the small entrepreneur in industrial and manufacturing sectors from.. urban areas”⁶¹ due to it attracting large investments and therefore larger companies. The metro overall attracts service sector big businesses, displacing and relocating smaller businesses in the industrial and manufacturing sector. The displacement of industrial and manufacturing sector drives off low-skilled workers from the area. Residential displacement due to unaffordable housing is also a possibility, because the metro is “likely to affect the property prices in undeveloped areas.”⁶² Less poor people will be able to live around the metro, further causing spatio-economic inequalities.

Potential Future Funding Issues

Hybrid and plug-in cars present potential future challenges for transportation infrastructure funding. Electric and hybrid cars need little to no gas, meaning the tax on gasoline at the state and federal level will cease to create revenue in the future. State and federal governments have already seen drops in revenue from gasoline taxes due to more fuel efficient cars on the road.⁶³ This has called for other funding sources to be implemented, including replacing the fuel tax with a fee on vehicle miles traveled, increasing the federal fuel tax, and taxing plug-in vehicles. Several states, including Colorado and Georgia, have implemented a

⁶¹ ibid

⁶² ibid

⁶³ (Dumortier, Kent, & Payton, 2016)

charge per-vehicle registration fee for alternative fuel vehicles ranging from \$50-\$200.⁶⁴

Washington is an example of a state that has implemented an alternative fuel vehicle fee of \$100.

The Washington governor's budget office estimated that the \$100 fee would bring in \$1.9 million for the 2015-2017 budget cycle.⁶⁵

A study to determine the impact of a tax of plug-in vehicles for transportation infrastructure funding found that "an increase in the federal fuel tax rate would generate significantly more revenue than the tax on plug-in vehicles" (Dumortier, Kent, & Payton, 2016). Currently, plug-in vehicles are a small portion of the car market, and do not significantly impact gasoline tax revenue. However, in the future the authors recommend a tax on the amount of miles a vehicle has traveled.

A simple increase in the federal excise tax on fuel has its potential draw backs. Basic economic principles would suggest that a raise on the gas tax could lead to an increase in consumers driving more fuel efficient cars. If gas costs more, people will want to find cost-effective ways to travel. Therefore, a raise on the gas tax could lead to a push for more fuel efficient and plug-in cars.⁶⁶ It has even been suggested that, because more affluent people tend to purchase fuel efficient cars, and lower income people tend to purchase cheaper, less fuel efficient cars, that a fuel tax increase would have negative impacts on lower income individuals. Research shows that the tax credits for plug-in vehicles are disproportionately used by higher income households.⁶⁷

Vehicle Miles Traveled tax (VMT) is an alternative that some states are considering

⁶⁴ (Dumortier, Kent, & Payton, 2016)

⁶⁵ (Dumortier, Kent, & Payton, 2016)

⁶⁶ (Mann, 2016)

⁶⁷ *ibid*

implementing to replace waning gas excise tax revenues. This would require a government agency to keep track of how many miles an individual has traveled, either through GPS, odometer readings, or self-reporting. The VMT tax is considered “fair” as an individual is only taxed on how far they drive, and those tax dollars will go towards transportation infrastructure. In turn, plug-in cars and fuel efficient cars will pay their “fair share” into transportation infrastructure funds.

A VMT tax has potential negatives as well. VMT administration and oversight causes cost concerns and privacy concerns. A VMT tax would cost money to collect; estimates for collecting VMT taxes range from 5% to 50% of the tax revenues. Germany reported that collected VMT tax on trucks on the autobahn costs more than 13% of tax collections.⁶⁸ Privacy concerns from VMT tax collection stem from the methods in which the government would assess the tax. GPS reporting of how many vehicle miles traveled is “...more accurate and less subject to fraud, (but) it reports to the government not only how many miles were traveled, but where and what time those miles were traveled.”⁶⁹ GPS reporting would be the most accurate, however it would also be the most invasive way to assess VMT taxes.

Increases in tolling have been discussed, as it is a form of a user fee and is generally seen as fair; people who do not use the road will not be subject to its maintenance. Increasing toll fees would face opposition from the trucking industry, and may cause adverse effects on traffic patterns (more diversion to free roads). Toll roads currently are a small portion of overall highway finance, and revenue projections from tolls would be uncertain.⁷⁰ Public-private

⁶⁸ *ibid*

⁶⁹ *ibid*

⁷⁰ (Mann, 2016)

partnerships are also seen as an alternative for future transportation infrastructure funding, as it lessens the tax burden on government. However, as before mentioned, these partnerships can garner controversy.

Summary

There is a wide array of literature on the economic impacts and importance that transportation infrastructure has on an economy. Transportation infrastructure can contribute directly to the wealth creation process, by being perceived as an additional input in the production process. Studies in Africa concluded that many countries on the continent lacked enough infrastructure to support their required levels of production. A study in China showed that transportation infrastructure investments have significant impacts on economic growth.

Transportation infrastructure keeps cities competitive, while a well-maintained infrastructure can convey a lot to the private sector. Well-maintained infrastructure can attract real-estate developers, while neglected transportation infrastructure can do just the opposite. The transportation infrastructure of a state also contributes to its competition by impacting its credit rating. It has been found that credit rating and status of transportation infrastructure has a positive correlation; the better the state of infrastructure the higher the credit rating. This is due to the fact that the status of infrastructure reflects management skills, financial strength, as well as the potential economic growth of a jurisdiction.

Subsequently, the neglect of transportation infrastructure has real consequences. The American Society of Civil Engineers found that a neglect of transportation infrastructure can lead to increases in operating costs for trucks, cars, and rail vehicles costs to damage to vehicles from deteriorated roadway surfaces, imposition to both additional miles traveled, time expended

to avoid unusable or heavily congested roadways, or due to the breakdown of transit vehicles, and an added cost of repairing facilities after they have deteriorated.”⁷¹

Keeping transportation infrastructure public vs. private is a very complex and controversial issue. Some economists have called for the total-privatization of transportation infrastructure, most notably roads and highways, and have laid out scenarios to prove that it is possible to implement. The rationale behind this is that market forces will incentivize “road entrepreneurs” to keep transportation infrastructure competitive and efficient. Public-private partnerships are another form of privatization which entails the government leasing roads to the private sector for a certain amount of time, usually over 20 years. The latter has happened in a number of U.S. states and has been suggested in Pennsylvania. In both cases, the perceived benefits include a lessened tax burden on the public and a new source of revenue for the government. Anticipated negative impacts of the privatization of roads and highways include a loss of revenue sources for the government (through the loss of toll revenue), less safe roads, high tolls prices and private control of transportation policy.

Transportation infrastructure funding will be a major concern for the future. As cars become more fuel efficient and less reliant on gasoline, tax revenue will plummet. While this is not an immediate concern, as the market share of plug-in vehicles increases, policy alternatives to excise taxes on gasoline will need to be explored. Among these alternatives is the most feasible, and already implemented, “Vehicle Miles Traveled” tax. This tax works as a direct excise tax on how much an individual uses a road, by tracking the amount of miles per year that vehicle drives. This yields controversy, due to the method which the tax would be assessed and

⁷¹ (American Society of Civil Engineers , 2014)

the implications of taxing per mile traveled. Green infrastructure will also be a subject that government will need to tackle. Green infrastructure seeks to lessen the pollution that hard-surface roads can produce by absorbing and filtering pollutants.

Although the positive impacts of transportation infrastructure have been widely examined, there seems to be adverse impacts as well. A study conducted on the Istanbul metro found that some transportation infrastructure can “...form segments and pathways; enclaves and fragments which may create isolation and/or connectivity and fluidity.”⁷² This is because the investment spurred by the Istanbul metro drove out small businesses and jobs for low-skilled workers, and the metro affects property prices in undeveloped areas, further causing displacement of poorer individuals.

Chapter IV: Data and Analysis

What is the impact that transportation infrastructure has on economies?

A healthy system of transportation infrastructure has a positive impact on an economy, be it at the local, state, or national level. The individual positive elements that transportation infrastructure yield include a better credit rating, greater real estate development, an increase in marginal production, private investment growth and cheaper maintenance costs for vehicles, all of which contribute towards two things:

1. The economy of a polity grows more competitive, making it more attractive to private sector investment; and

⁷² (Beyazit, 2015)

2. The economy grows due to a growth in productivity, especially in countries or areas with low stocks of public capital.

The first observation, growth in competitiveness and private sector investment, may contribute to the second observation, a growth in productivity. When the cost of doing business is easier, cheaper, and safer in areas with developed and reliable transportation infrastructure, businesses will prefer those areas; this is a basic example of economic competition. However, in this case, the phenomenon is occurring between actors in the public sector as opposed to private firms.

As demonstrated with the Istanbul metro, socio-economic inequalities can occur when transportation infrastructure is introduced into an area. However, I hypothesize that these inequalities are exacerbated in areas or countries with low stocks of public capital, or in countries or areas with heavy reliance on the industrial or manufacturing sector. In Pennsylvania, there are planning organizations to address this problem and ensure that lower-income individuals have fair access to transportation as well.⁷³

Transportation infrastructure investment in Pennsylvania would most likely contribute more towards its competitiveness and, in turn, to private sector investment as opposed to a drastic increase in productivity due to the fact that the state already has a high stock of public capital. If Pennsylvania has a better quality of transportation infrastructure over its neighbors, it will be more economically competitive compared to the other states. This in turn will lead to an increase in private sector investment and growth. The second observation, a direct increase in productivity, may hold true in local economies. especially in more isolated or rural portions of the Pennsylvania. Local economies with better access to large-scale economies through

⁷³ (Meredith Biggica, 2017)

improved transportation infrastructure, might experience a more drastic increase in productivity, depending on the amount of public capital in the area. Efforts by the Maintenance and Preservation program to improve low-volume roads, mainly in rural areas, could have positive impacts on these economies.

In both rural and urban areas the quality of transportation infrastructure would be improved, and the biggest and most immediate impact would be a lowered cost of maintenance to vehicles driving in those areas. The ASCE stated that the costs associated with defunct transportation infrastructure would cause business to run inefficiently,⁷⁴ which means the opposite would hold true for efficient transportation infrastructure. Lower vehicle maintenance costs will lead to lower costs to transport private capital, and contribute towards businesses running more efficiently.

Applying Dr. Agenor's findings

Dr. Agenor stated that public capital has a direct productivity and cost effect on private production inputs. Research has found that well-maintained transportation infrastructure would lower transportation costs, therefore increasing private productivity. This has been studied by other academics conducting research on the impact of transportation infrastructure; Africa, China and the U.S.A.⁷⁵ have and will experience productivity and cost effects on private production inputs as transportation infrastructure improves. But as Dr. Agenor has stated, this effect is larger in countries with smaller initial stocks of public capital.⁷⁶

⁷⁴ (American Society of Civil Engineers, 2016)

⁷⁵ (Jiang, He, Zhang, Qin, & Shao, 2017); (Kodongo & Ojah, 2016); (American Society of Civil Engineers, 2016)

⁷⁶ (Agenor, 2013)

Dr. Agenor also observed that public capital has a complementarity effect on private investment. This is something that my research has confirmed. Private businesses will choose locations with better transportation infrastructure, as transportation costs will be lower, and labor productivity will be higher among other occurrences. This is apparent in the increase in real estate development that ULI noted,⁷⁷ as well as in the case of the Istanbul metro.⁷⁸

Dr. Agenor's third observation that public capital investment could lead to a crowding-out effect on private spending through the financial system, varies depending upon the way transportation infrastructure is paid for. Research has shown that the quality of transportation infrastructure is correlated with credit score.⁷⁹ The quality of transportation infrastructure can convey a states' financial management among other things, which is what factors into its credit score. Transportation infrastructure can therefore convey the same things to private sector firms, and may show for a states' financial situation.

Pennsylvania has the highest gas tax rate in the nation, with 58.20 cents per gallon as of January 2017.⁸⁰ This high gas tax, a consequence of Act 89 of 2013, may lead to a decrease in Pennsylvania's competitiveness if the tax rate stays that high. As the gas tax is an indirect tax on businesses (unless that business is explicitly tied to transportation) it alone may not trigger a crowding-out effect. Analysts will need to see what impact this rise in taxes will have on economic growth in the short term as well as the long term.

How other states pay for Transportation infrastructure

⁷⁷ (Urban Land Institute , 2014)

⁷⁸ (Beyazit, 2015)

⁷⁹ (Chen, A, & Wang, 2016)

⁸⁰ (Drenkard, 2017)

Since 2012, twenty-four states have approved plans to raise additional transportation revenues, similar to Act 89. These states include some of Pennsylvania's neighbors; New Jersey, Delaware, Maryland, and Ohio.⁸¹ States have raised funds for transportation infrastructure through a number of means, including:

- Increases in tolls and increasing the number of toll roads;
- Issuing general obligation bonds;
- Increasing license fees, and adding fees for hybrids and electric cars;
- Increasing fuel excise taxes;
- Fees for short-term lodging;
- Allowing counties and groups of counties to impose higher sales taxes;
- Allowing transit systems to spend more of its locally-raised revenue to fund operations;
- Dedicating excess dollars from the states' general fund;
- Dedicating other state funds to transportation infrastructure;
- Increasing taxes on tobacco products; and
- Allowing governments and government agencies to accrue more debt.

Almost all states are increasing gas excise taxes, many are increasing toll fees and vehicle licensing/registration fees, and some are using sales tax revenue. Some states will hold referendums to raise sales or property tax rates, issue bonds, or increase fees.⁸² Pennsylvania is a state with one of the more comprehensive initiatives towards funding for transportation infrastructure. Pennsylvania is also a state that does not rely on or authorize more debt

⁸¹ (Transportation for America , 2016)

⁸² *ibid*

accumulation to fund its transportation infrastructure initiatives.

A state not included on the list, Colorado, has used revenue from marijuana sales tax to construct roads in the town of Pueblo West.⁸³ The Pueblo County Board of Commissioners voted on a resolution to allot a portion of tax revenue from marijuana sales. Across Colorado, small towns as well as large urban areas are benefitting from marijuana sales. These communities are using the revenue from marijuana taxation and regulation for road projects and improvements. In many cases, revenue from taxes on the sale of marijuana goes into the community's general fund.⁸⁴ In 2015, the city of Denver alone took in \$29 million from the taxing and regulation of marijuana markets. In Colorado, the ability for local communities to levy their own taxes on the marijuana industry is what makes the biggest impact on their budgets. Kris Teegardin, the mayor of Edgewater, Colorado, stated that "Tax sales have been able to speed up major infrastructure improvements that we would otherwise have to take a very incremental approach to."⁸⁵ However, tax analysts suggest that marijuana legalization may not be enough to solve big budgetary problems that states face.⁸⁶ It seems that marijuana legalization can have the biggest impact on local communities if those communities are able to levy their own taxes. The boom in tax money from marijuana taxation and regulation in states that have legalized it recreationally may also be the result of marijuana tourism, something that Colorado government officials are very aware of.

What is the most effective way to fund Pennsylvania's transportation infrastructure?

⁸³ (KOAA News 5, 2016)

⁸⁴ (Illescas, 2016)

⁸⁵ (Wight & Runyon, 2016)

⁸⁶ Ibid.

Through the research that has been conducted for this paper, to answer “What is the most effective way to fund Pennsylvania’s transportation infrastructure?” one must first ask if the states’ transportation infrastructure should remain within the public sector or be transferred to the private sector.

If Pennsylvania’s transportation infrastructure is to remain in the public sector, the funding should come from a number of sources. Act 89 of 2013 is one of the first concrete steps in state funding for transportation infrastructure. The plan eliminates the flat gas tax and uncaps the wholesale oil franchise tax as to offset and make the flat gas tax elimination revenue neutral. Over a five year phase out of the oil franchise tax is estimated to generate an additional \$1.85 billion per year. The Act also increases vehicle registration fees, driver licensing fees and traffic violation fines over the course of years. Although various sources have stated that Act 89 does not do enough to fill in the infrastructure funding gap in Pennsylvania, it sets ground for future endeavors. A problem with Act 89 is the gas tax rate it implements. Aforestated in the paper, Pennsylvania now has the highest gas tax in the Country.⁸⁷ While this alone is not a reason to be against Act 89, the externalities from increasing tax rates to this level should be taken into account.

The Road Maintenance and Preservation program is another program that should be expanded upon, as the program focuses on low-volume roads in rural parts of the Commonwealth. These are roads that have been neglected for years, and may not have been a priority in Act 89 projects. It would also be beneficial to capitalize the Pennsylvania Infrastructure Bank and other state and federal loaning organizations with more funds, so that the

⁸⁷ (Drenkard, 2017)

bank is able to lend out more money for infrastructure projects.

The federal government will need to step in for funding of state transportation infrastructure as well. Federally aided roads account for 28,221 miles in the state, and federal dollars are very important in the interstate system and National Highway system in Pennsylvania. As both presidential candidates understood large sums of money would be needed to repair the nation's infrastructure, it is important that the winner and current president, Donald Trump, acts upon his words. The various federal agencies that deal with transportation infrastructure funding such as the Federal Highway Administration and the Federal Railroad Administration will need to step up their funding to the states if the federal government truly understands the economic detriments that can occur from neglecting the nation's infrastructure.

Pennsylvania should choose less-volatile tax sources to fund transportation infrastructure, however this is a politically touchy subject. The further one moves away from excise taxes to fund transportation infrastructure, the harder it is to gain political and public support. Individuals who do not directly use mass transit, rail or roads may not see why they should be taxed for a service or good that they do not use. Using funds from the General fund of Pennsylvania for roads, etc. would contribute a more stable funding source for the state. Other states have used sales tax to either pay back government bonds used towards transportation infrastructure or to directly pay towards transportation infrastructure.⁸⁸ Even though some individuals may not directly use certain forms of transportation infrastructure, everyone receives some form of benefit from transportation infrastructure. If other states are able to raise non excise taxes, Pennsylvania should follow suit.

⁸⁸ (Transportation for America , 2016)

Raising taxes that directly impact businesses or taking out bonds can lead to a crowding-out effect by the private sector, as noted by Dr. Agenor. Distortionary taxes can directly affect a business' capital formation, which may lower incentives to move to an area. Large amounts of public debt due to bonds increase the risk of a default, which can have a negative effect on private capital accumulation and economic competition. This crowding-out effect tends to be short term, as the increase in public capital stock raises output growth in the medium and longer term, and will result in higher tax revenues.⁸⁹

Infrastructure privatization is another option that the state can pursue. The most obvious benefit of privatization is a decreased tax burden on the state. The state would not need to raise tax rates, accumulate more debt, or implement more fees. The state would receive profits from selling or leasing roads to private firms, and would be able to use that revenue for other purposes, such as paying debts or investing in other programs. Infrastructure privatization is the most politically unappealing choice, as it is difficult for many people to fathom how roads and other public assets could be privately owned, nor do people want to lose direct public control of transportation policy. Pennsylvania faced backlash from the idea of privatizing the turnpike just years back from members of the public as well as the trucking industry,⁹⁰ and there is no question that another proposal for infrastructure privatization would be met with fierce lobbying.

Chapter V

Policy Recommendations

There is a wide body of information on the economic impacts of transportation

⁸⁹ (Agenor, 2013)

⁹⁰ (Carretta, 2008)

infrastructure on a country. Transportation infrastructure is a crucial part of any polity, and its quality can greatly impact an economy. Given the research conducted on the topic, the following policy recommendations could be effective ways to handle transportation infrastructure for the future.

1). Expand taxation for the funding of transportation infrastructure to non-excise, less-volatile tax sources.

Raising excise taxes to pay for transportation infrastructure is a good start, however Pennsylvania must follow suit of other states and look towards other non-excise sources for tax revenue. States have implemented increases on sales taxes, taxes on short term lodging, taxes on tobacco and marijuana, increased fees for vehicle registration and have appropriated funds from a general fund or other fund. I hypothesize that the expansion of revenue for transportation infrastructure beyond excise taxes, without accumulating debt, will hamper the crowding-out effect that Dr. Agenor has identified. I would also recommend using less-volatile tax sources, so that transportation infrastructure receives a steady source of revenue. In Pennsylvania, the least volatile tax source is the sales tax.⁹¹

2). Take out bonds to pay for transportation infrastructure, and pay the debt back with less-volatile tax sources.

Several states have enacted referendums to issue millions of dollars in bonds to fund various transportation infrastructure projects. In Georgia, local government leaders will identify

⁹¹ (The PEW Charitable Trusts , 2017)

road, bridge, transit, rail, port and airport projects to be funded in their respective county, and will hold referendums to allow these counties to bond against the money required for the project. I believe that this method would work best for Pennsylvania, as it allows voters to decide how to pay for transportation infrastructure projects. Revenue to pay back these bonds would be raised through referendum approved increases in taxes; counties in Georgia would have the ability to raise the local sales tax up to 1%.⁹² This would be the most immediate policy option, however the crowding-out effect that Dr. Agenor identified would be more likely to occur with this alternative.

3). Privatization of Transportation infrastructure

As stated before, privatization is not the most attractive option, however the tax burden on the state would be significantly lightened. To Walter Block and other economists like him, economic competition would ensure that roads have an incentive to be efficient. Even former Governor Ed Rendell likened the idea of privatizing the Pennsylvania turnpike. I would recommend starting by privatizing or leasing the largest roads in the state, such as the turnpike, and use the profits from its sale for other road projects across the state. The state could make at least \$12.8 billion from selling the turnpike, going by the offer from 2008. Former Governor Rendell stated that the firm that would have leased the road would have repaired and upgraded the road, as well as had to meet strict quality standards imposed by the commonwealth. Rendell also assured the public that "...that lease payment would be 13% higher than.. that made from tolling. Leasing the road will limit toll increases, protect the turnpike workers' collective

⁹² (Transportation for America , 2016)

bargaining agreement, ensure scheduled turnpike improvements are completed and maintain the condition of the turnpike..”⁹³ This would solve some of the concerns that the U.S. PIRG illustrated, such as the state not getting a fair price for their roads, and that private control of roads could lead to a decline in road safety and maintenance standards.

Privatization of other transportation infrastructures has already been implemented, such as rail freight transport. It is possible for other transportation infrastructures, such as commuter rail and bus, to be privatized and to have firms work in cooperation with the state. It would be possible for local polities such as townships, cities and countries to privatize or lease some or all of their road systems. Starting on a small scale would allow us to understand what method works the best and is the most efficient. The states act as laboratories of democracy, and trying new things such as transportation infrastructure privatization in Pennsylvania would act as an economic experiment.

Policy trends for the future

In the future, I would not recommend implementing a Vehicle Miles Traveled tax as some other states have experimented with. The biggest problem I see with the implementation of VMT is the method through which the tax is calculated. The most accurate way for the government to calculate a VMT tax for an individual is to assess how many miles that person had driven for a year. This would require a GPS reporting, which is intrusive to people’s lives. GPS tracking would tell the government not only how many miles a person traveled, but also where and what time those miles were traveled. In Oregon where the VMT taxation program has

⁹³ (Carretta, 2008)

been tested, “protection of privacy proved to be the most prominent concern of the citizenry...”⁹⁴

Another potential problem I see with VMT is the cost it takes to assess and collect the tax.

Estimates range from 5%-50% of tax revenue would need to be used to actually assess and collect the tax.⁹⁵

Once electric vehicles become a larger share of the cars on the road, to the point where electric cars significantly affect gas tax revenue, then a fee or tax on electric cars may be necessary. Taxes on Car Charging Stations, where electric cars can get a quick charge while on the road, would be a realistic solution for revenue as gas stations become less prominent. Tax sources for transportation infrastructure in the future will need to evolve as transportation evolves.

Gaps in the literature

While there is a large amount of information on the effects of transportation infrastructure, there are still gaps in the literature that should be addressed. The “public vs. private” argument brings up an important question that should be explored further: Does public investment or private investment in transportation infrastructure more significantly impact productivity growth and competitiveness? By answering this question, policy makers would better know when to use government intervention as a solution or let market forces work. Policy makers should also question what should be defined as a public good and what should not be. If a product or service that has historically been under control of the public sector can more effectively be provided by the private sector, what is stopping that public good from becoming a

⁹⁴ (Mann, 2016)

⁹⁵ (Mann, 2016)

private good?

Conclusion

By understanding the economic impacts that transportation infrastructure has on a polity, policy makers, politicians, and members of the public will better be able to approach issues regarding it in the future. The vast economic benefits derived from adequate transportation infrastructure range from spurring private investment to direct and indirect impacts on the growth of economic productivity of a region. Identifying effective and stable ways to fund transportation infrastructure will ensure that healthy and effective infrastructure is maintained for the future. I have no doubt that most individuals understand that transportation infrastructure is important for a society, however many would not understand exactly in what ways it benefits a society. By utilizing this information, state and federal policy makers and legislators can take the appropriate steps to stimulate economic growth through transportation infrastructure development in Pennsylvania.

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